Article

Predicting Passenger Car Prices with Machine Learning Models

Burmachach Nuradilova[1,†,‡], Remudin Reshid Mekuria[1,†,‡,*]

[1] Applied Mathematics and Informatics Department, Ala-Too International University, Bishkek, Kyrgyzstan
[*] Correspondence: remudin@alatoo.edu.kg
[†] Current address: Ankara 1/8, Bishkek
[‡] These authors contributed equally to this work.

Abstract: The global automotive market has shown consistent long-term growth, with periodic declines linked to global crises. Kyrgyzstan mirrors this trend: in 2023, the number of registered passenger cars increased by 24 % compared to 2016, reflecting rising private car ownership and expanding market activity. As the local car market continues to grow, accurate car price prediction becomes increasingly important for both consumers and sellers. This study introduces a data-driven model to estimate car prices using real-world listings from one of the largest automotive marketplace in Kyrgyzstan. The dataset is user-generated, capturing authentic market behavior alongside the inherent inconsistencies typical of manually submitted entries. Through exploratory data analysis and visualization, we investigated the structure of the local market, identified key pricing factors, and examined prevalent trends in vehicle listings. After comprehensive preprocessing, which included data cleaning, transformation, and feature selection, we trained and evaluated multiple machine learning models. Among these, CatBoost achieved the highest performance, with a Mean Absolute Percentage Error (MAPE) of 9.61%, confirming its suitability for this task. This research demonstrates how combining exploratory data analysis with advanced machine learning techniques can reveal valuable insights and enhance pricing accuracy and transparency in the automotive market.

Keywords: Machine learning, CatBoost regressor, Price prediction, Web scraping, Feature engineering, Data preprocessing, Regression models, Model evaluation, Car price estimation, Kyrgyzstan car market

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0. Introduction

This study not only contributes to the growing body of literature on machine learning applications [1] in price forecasting but also addresses a significant gap in region-specific automotive market analysis. Unlike global vehicle pricing models [2] that often overlook local trends and consumer behavior, our research contextualizes price prediction within Kyrgyzstan's unique market dynamics, such as import patterns, registration nuances, and shifting fuel preferences.

In conducting this research, we also address the challenges posed by noisy and unstructured advertisement data, which is manually entered by sellers on the platform [3]. This introduces inconsistencies and errors that require careful preprocessing, including missing value handling [4], normalization [5], and categorical encoding [6]. By engineering relevant features and rigorously evaluating model performance through metrics such as Root Mean Squared Error (RMSE) and R^2 score, we aim to develop a predictive framework that balances accuracy with practical usability.

The insights generated from this model have potential applications beyond academic inquiry. Accurate car price prediction tools can enhance transparency on platforms like mashina.kg, improve buyer confidence, help sellers competitively price vehicles, and inform automotive financing or insurance assessments. Furthermore, the proposed methodology can serve as a template for similar data-driven studies in other emerging markets.

This research demonstrates how the integration of machine learning and data-driven exploration can enhance transparency in the Kyrgyz automotive market and support smarter pricing decisions. The remainder of this paper is structured as follows: Section 1 reviews related literature, Section 2 details the methods and tools used, Section 3 presents and discusses the results, and Section 5 concludes the study with final insights and directions for future research.

1. Literature Review

The prediction of used car prices has seen substantial evolution, advancing from basic linear models to sophisticated hybrid machine learning approaches. Early efforts primarily relied on simple models trained on small, manually collected datasets [7]. One such study applied techniques such as k-nearest neighbors, decision trees, multiple linear regression, and Naïve Bayes to estimate vehicle prices from newspaper listings in Mauritius. Despite using a broad set of features—including brand, engine size, transmission type, and mileage—these models achieved limited accuracy, often below 70%, due to dataset limitations and difficulties in processing numeric values.[8]

As more structured datasets became publicly available, researchers turned to traditional statistical models. A study utilizing Ordinary Least Squares (OLS) regression on a Kaggle dataset reported a high R^2 score of 0.90. However, the model exhibited sensitivity to extreme values, particularly in underrepresented price segments, indicating that linear models may lack the flexibility to capture the full range of market behaviors.[9]

Subsequent studies embraced non-linear models and ensemble methods. One example [10] employed support vector machines (SVM), XGBoost, and neural networks on a large-scale dataset of 200,000 vehicle listings. After preprocessing and dimensionality reduction using PCA and ISOMAP, XGBoost achieved the best performance with an R^2 of 0.982, demonstrating the effectiveness of boosting algorithms in capturing complex price dependencies.

A more advanced approach integrated Grey Relational Analysis (GRA) and Particle Swarm Optimization (PSO) with a Backpropagation Neural Network (BPNN), forming a hybrid PSO-GRA-BPNN model [11]. This architecture achieved a MAPE of 3.94% and $R^2 = 0.984$, outperforming traditional regression and standalone machine learning models. However, the increased complexity and training time posed scalability challenges for practical applications.

In contrast to previous studies based on curated datasets, the present work uses real-world data collected from *mashina.kg* [3]. The data is user-generated, reflecting authentic market behavior along with the inherent inconsistencies typical of manually entered listings. Without relying on hybrid architectures or discretization schemes, we apply a single CatBoost model to predict car prices directly. While our model's MAPE of 9.61% is modest compared to the most optimized models, it balances accuracy and simplicity, making it suitable for practical deployment and for capturing region-specific pricing dynamics.

2. Method and Tools

2.1. Data Collection

The automotive market in Kyrgyzstan has experienced substantial growth in recent years, with a notable 24% increase in the number of registered passenger cars in 2023 compared to 2016 [12]. This growth reflects the rising trend of private car ownership and the expanding scope of the local market. The data used in this study was obtained through a custom web scraping process from *mashina.kg*, a widely used car advertisement platform in Kyrgyzstan. The resulting dataset consisted of 31,639 entries across 35 columns, capturing key vehicle attributes such as brand, model, mileage, year of manufacture, color, and body type [13]. Furthermore, the data set included several textual description fields detailing specific features, such as air conditioning, MP3 players, and airbags, as well as attributes related to exterior design, interior materials, multimedia systems, safety features, and optional equipment.

2.2. Data Cleaning

An exploratory data analysis was first conducted to understand relationships between variables and to identify issues such as missing values, outliers, and data imbalance. The full dataset contained 46 distinct car brands, where each brand represents a manufacturer (e.g., Toyota), and all models under that brand were grouped accordingly. To reduce noise and focus on statistically relevant data, brands with fewer than 20 entries were excluded. This filtering step resulted in a refined dataset of 29 brands, providing a more reliable foundation for modeling and pattern discovery.

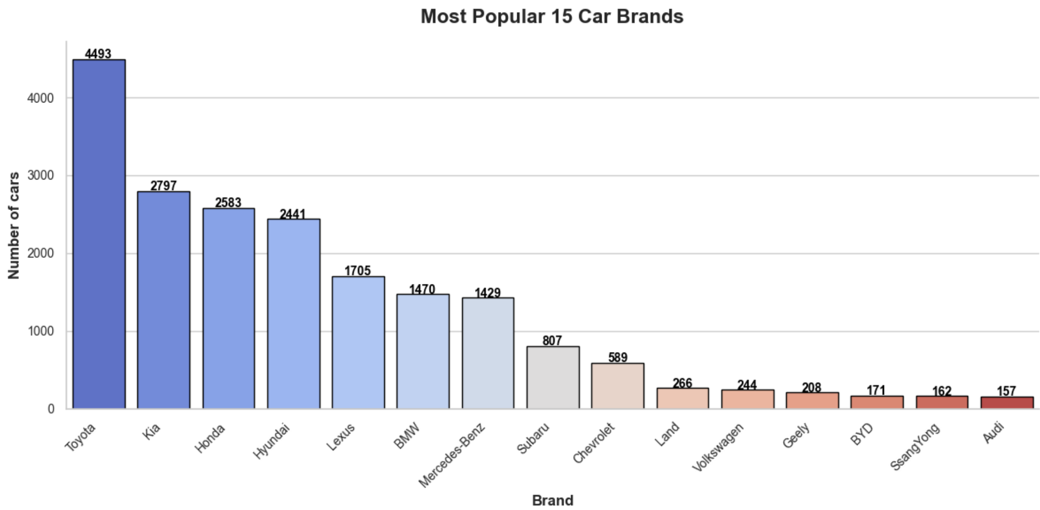

Figure 1. Most Popular 15 Car Brands

Figure 4 illustrates the distribution of the 15 most frequently listed brands, highlighting Toyota, Kia, and Honda as the most dominant entries in the Kyrgyz car advertisement market.

To address missing values, appropriate imputation strategies were applied to preserve data integrity and minimize information loss. For example, missing mileage values were imputed using group-wise mean based on brand, model, and year to ensure contextual accuracy.

Following the imputation process, the dataset was further analyzed to reveal underlying patterns and inconsistencies. A key part of this exploration involved examining the distribution of average vehicle prices across different availability statuses and condition levels. This analysis is visualized in the heatmap shown in Figure 2, which highlights significant price differences between cars listed as *in stock*, *on order*, and *in transit*.

Figure 2. Average Price by Car Availability and Condition

The heatmap analysis revealed that vehicles listed as *in stock* generally exhibited lower average prices compared to those categorized as *on order* or *in transit*. Based on this observation—and to align with typical resale market behavior—only **in-stock** vehicles were retained for model training. This filtering step ensured that the final dataset more accurately reflected prevailing market dynamics.

2.3. Feature Engineering

The feature engineering process [14] focused on constructing intuitive and informative variables to enhance model performance. Based on the advertisement posting date, temporal features were derived: *Seasonality*, indicating the season in which a vehicle was listed, and *Days Ago*, representing the number of days elapsed since the listing was published.

To better capture vehicle usage patterns, a new variable—*Mileage per Year*—was created by dividing the total mileage by the car's age in years. This provided a normalized measure of annual usage, offering greater insight into the vehicle's condition beyond raw mileage alone. Additionally, engine volume values were standardized to accommodate variations in measurement units across fuel types (e.g., liters for internal combustion engines vs. kilowatts for electric vehicles), ensuring consistency and comparability throughout the dataset.

Several non-informative or incomplete features were excluded during preprocessing. Fields such as *url*, *added_time*, and *car_availability* were removed, as only in-stock cars were retained for analysis, and the remaining fields offered no predictive value. The *VIN* (Vehicle Identification Number) column was also dropped due to its limited usability—only the first three characters were visible, and more than 80% of entries contained missing values, making it unsuitable for modeling.

2.4. Model Implementation

To train and evaluate the predictive models, the preprocessed dataset was split into training and test sets using an 80/20 ratio [15]. This approach ensured that the models were evaluated on unseen data, allowing for a realistic assessment of generalization performance.[16]

Given that the target variable, car price, spans a wide numerical range ($5,000 to $120,000), a logarithmic transformation [17] was applied to normalize the distribution. This helped stabilize variance, reduce the impact of outliers, and improve model learning for high-price vehicles. All models were trained to predict the natural logarithm of price, and predicted values were later exponentiated to obtain results in the original scale.

Four different regression algorithms were implemented for model comparison:

- Random Forest Regressor (RF)
- Gradient Boosting Regressor (GBR)
- Partial Least Squares Regression (PLSRegression)
- CatBoost Regressor

For models that do not natively support categorical features, namely Random Forest, Gradient Boosting, and PLSRegression categorical variables were encoded using *LabelEncoder*. This encoding approach was selected for its simplicity and compatibility with tree-based models, which are generally insensitive to ordinal relationships in encoded values. CatBoost, in contrast, natively supports categorical variables, allowing it to handle such features without preprocessing.[18]

All models were trained using default hyperparameters initially, with further tuning performed based on cross-validation performance. Evaluation metrics are reported in the following section.

2.5. Model Evaluation

The results from the model evaluation and hyperparameter tuning phases [19] indicate that CatBoost consistently outperformed the other models across all key metrics, particularly in Mean Absolute Percentage Error (MAPE) and R^2 score. With a MAPE of 9.61% and R^2 of 0.9448, the model demonstrates a high level of predictive accuracy on unseen data.

These outcomes suggest that CatBoost's native support for categorical variables, combined with its gradient boosting framework, offers a distinct advantage in modeling complex, high-dimensional relationships. In contrast, models such as PLSRegression and DecisionTreeRegressor struggled to capture non-linear interactions and hierarchical feature structures, resulting in noticeably lower R^2 scores and higher error rates, as shown in Table 1.

Model	MAE	RMSE	R^2	MAPE
CatBoost	**1970.20**	**3762.59**	**0.9448**	**9.61%**
Random Forest	2860.53	4507.42	0.9199	11.82%
DecisionTreeRegressor	2860.53	5835.22	0.8708	14.32%
GradientBoostingRegressor	3291.16	5632.97	0.8796	18.35%
PLSRegression	3955.81	6538.53	0.8378	25.98%

Table 1. Performance comparison of different regression models.

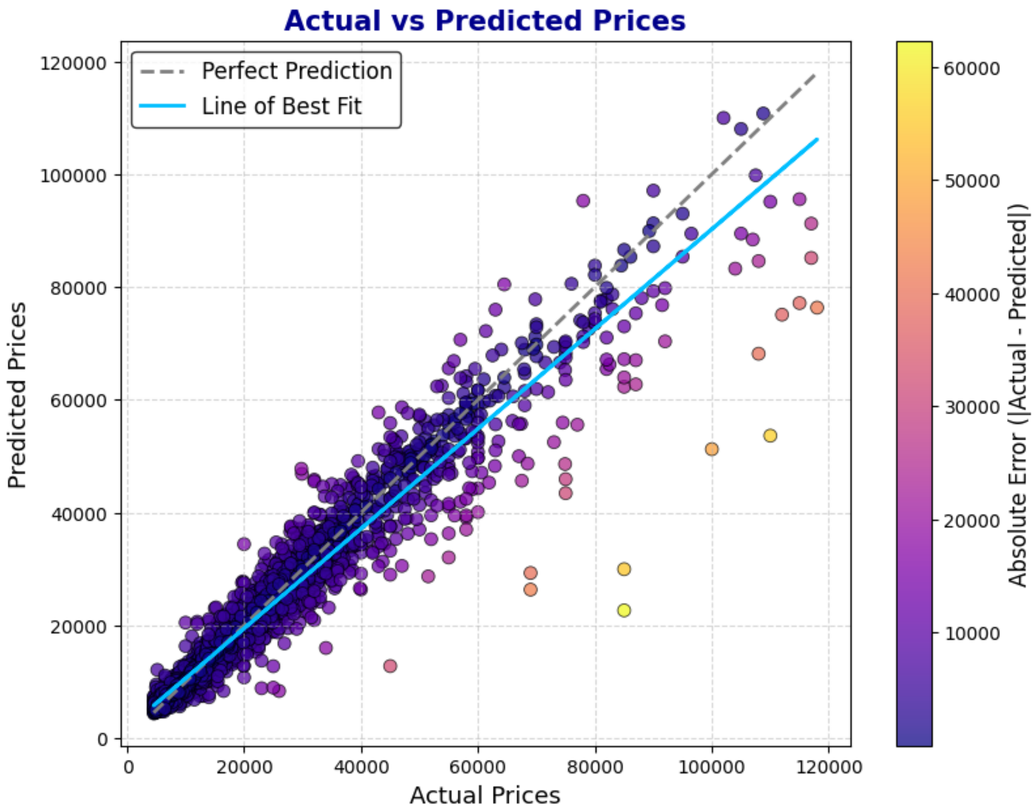

Figure 3. Comparison Between Predicted and Actual Vehicle Prices

Figure 3 presents the predicted versus actual car prices for the CatBoost model. The points are tightly clustered around the diagonal line, indicating strong predictive alignment across a wide range of vehicle prices.

Despite CatBoost's robust performance, some residual errors persist, particularly for vehicles at the upper and lower ends of the price spectrum. These discrepancies may be attributed to listing inaccuracies, rare combinations of features, or non-standard pricing strategies employed by sellers.

In summary, the results validate CatBoost as a highly effective model for predicting vehicle prices in a real-world, user-generated dataset. Its ability to model complex patterns with minimal preprocessing makes it well-suited for deployment in online car marketplaces, where dynamic pricing tools and intelligent price suggestions can provide immediate value to both sellers and buyers.

2.6. Hyperparameter Tuning

To enhance predictive performance, hyperparameter tuning was conducted using `Optuna` [20], a modern framework for efficient hyperparameter optimization. Cross-validation was applied across all data subgroups during tuning to ensure robustness and to mitigate overfitting to specific segments of the training set. This strategy enabled a more reliable assessment of each model's generalization capability.

Through this process, `Optuna` identified the optimal set of hyperparameters for the CatBoost model, as detailed in Table 2.

Hyperparameter	Description	*Value*
iterations	Number of boosting rounds	*1000*
depth	Maximum tree depth	*8*
learning_rate	Learning rate for updates	*0.1*
l2_leaf_reg	L2 regularization coefficient	*10*
bagging_temperature	Randomness in data sampling	*0.7944*
border_count	Binning size for continuous features	*50*

Table 2. Optimized hyperparameters for CatBoost using Optuna

These optimized settings enabled the CatBoost model to better capture non-linear feature interactions and subtle patterns in the data, ultimately resulting in the highest predictive performance among all models tested.

3. Results

3.1. Overall Performance

The CatBoost model was evaluated using multiple performance metrics on both training and test datasets. These include Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 Score, Median Absolute Error, and Explained Variance Score. The results are summarized in Table 3.

Metric	**Train**	**Test**
Mean Absolute Percentage Error (MAPE)	**7.53%**	**9.61%**
Mean Absolute Error (MAE)	1494.25	1970.20
Root Mean Squared Error (RMSE)	2836.98	3762.59
R^2 *Score*	0.9679	0.9448
Median Absolute Error	755.58	941.30
Explained Variance Score	0.9680	0.9452

Table 3. Performance metrics for training and testing data.

3.2. Performance by Price Range

To evaluate the model's robustness across various market segments, the test data was divided into price bins ranging from \$4.5k to \$120k. As shown in Table 4, the CatBoost model demonstrates strong predictive performance within the \$4.5k–\$60k range, achieving MAPE values below 11%. However, performance begins to degrade for higher price segments, with MAPE increasing to over 15% for cars priced between \$100k and \$120k.

Price Range	**MAE**	**RMSE**	**MAPE**
4.5K–10K	695.40	1028.24	10.22%
10K–20K	1201.66	1703.48	8.31%
20K–40K	2837.90	4023.82	10.17%
40K–60K	5295.70	6998.24	10.73%
60K–80K	8861.74	10663.24	13.00%
80K–100K	12744.80	16300.29	14.66%
100K–120K	16653.15	18216.80	15.30%

Table 4. Performance metrics across different price ranges.

The variation in performance closely mirrors the distribution of data across price bins [21] . Figure 4 shows that the dataset is highly imbalanced, with the vast majority of listings falling below the \$40k mark. The \$10k–\$20k bin alone accounts for over 8,000 vehicles, while fewer than 50 listings fall in the \$100k–\$120k range. This imbalance significantly limits the model's ability to learn reliable patterns in higher price segments, which explains the increased prediction error.

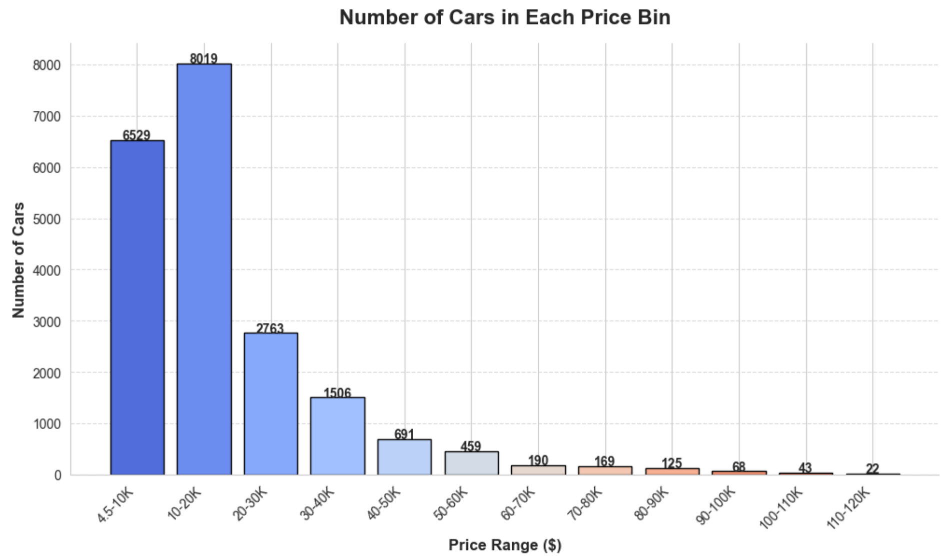

Figure 4. Distribution of cars across price ranges

This trend is further supported by the residual plot in Figure 5, which shows increasing prediction variance at higher price levels. It illustrates the residuals [22] (actual minus predicted prices) plotted against the actual vehicle prices. While the model performs reliably within the low to mid-price range, a visible spread of residuals emerges as prices increase. This pattern confirms that the prediction error grows with price, likely due to the lower density of training data for high-end vehicles. The asymmetry and widening dispersion indicate that the model tends to slightly underpredict expensive vehicles, further reinforcing the impact of data imbalance on prediction quality.

Figure 5. Error distribution between actual and predicted prices. Errors less than 5% are shown in blue, while larger errors are highlighted in red to indicate more significant deviations.

Together, these results indicate that the model performs best on mid-range vehicles—the most prevalent class in the dataset—while performance deteriorates in underrepresented high-end segments due to data sparsity.

3.3. Performance by Brand

To further examine brand-specific model performance, a bubble plot was constructed, as shown in Figure 6. The x-axis represents the number of car listings per brand, while the y-axis indicates the average prediction error, defined as the difference between actual and predicted prices. Each bubble's size is proportional to the volume of listings for that brand, providing a visual sense of brand representation in the dataset.

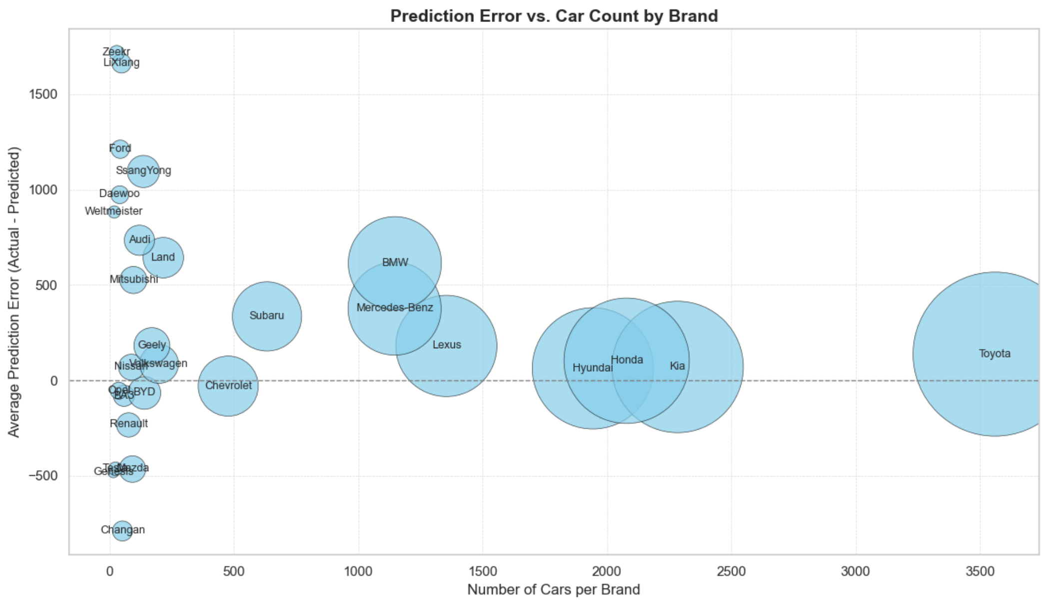

Figure 6. Bubble plot showing brand-wise prediction error vs. listing count. Bubble size represents the number of car listings per brand. Brands with fewer listings tend to exhibit higher and more variable prediction errors.

Brands with a high number of listings, such as Toyota, Kia, Hyundai, and Honda, are located on the right side of the plot and show minimal average error, indicating that the model performs reliably for these well-represented brands. In contrast, brands with limited representation (e.g., Zeekr, Lixiang, Changan) are clustered on the left and exhibit significantly higher and more variable prediction errors. This trend underscores the effect of data imbalance: the fewer examples the model sees, the less accurately it can learn brand-specific pricing patterns. Additionally, premium brands like BMW and Lexus exhibit mild underprediction, likely due to pricing complexity not fully captured by the model. Overall, this visualization reinforces the need for brand-aware modeling strategies or sampling adjustments in future work.

3.4. Feature Importance

To better understand the underlying drivers of vehicle price prediction, we extracted feature importance values [23] from the trained CatBoost model. These scores reflect the relative influence of each input variable on the final output, based on their frequency and impact in the model's decision-making process. Figure 7 presents the top 20 most important features, as ranked by CatBoost. Several key insights can be drawn:

- **Age:** Age (inverse of Year of Manufacture) emerged as the most significant predictor. This aligns with market expectations, as newer vehicles typically retain more value and are associated with lower depreciation.

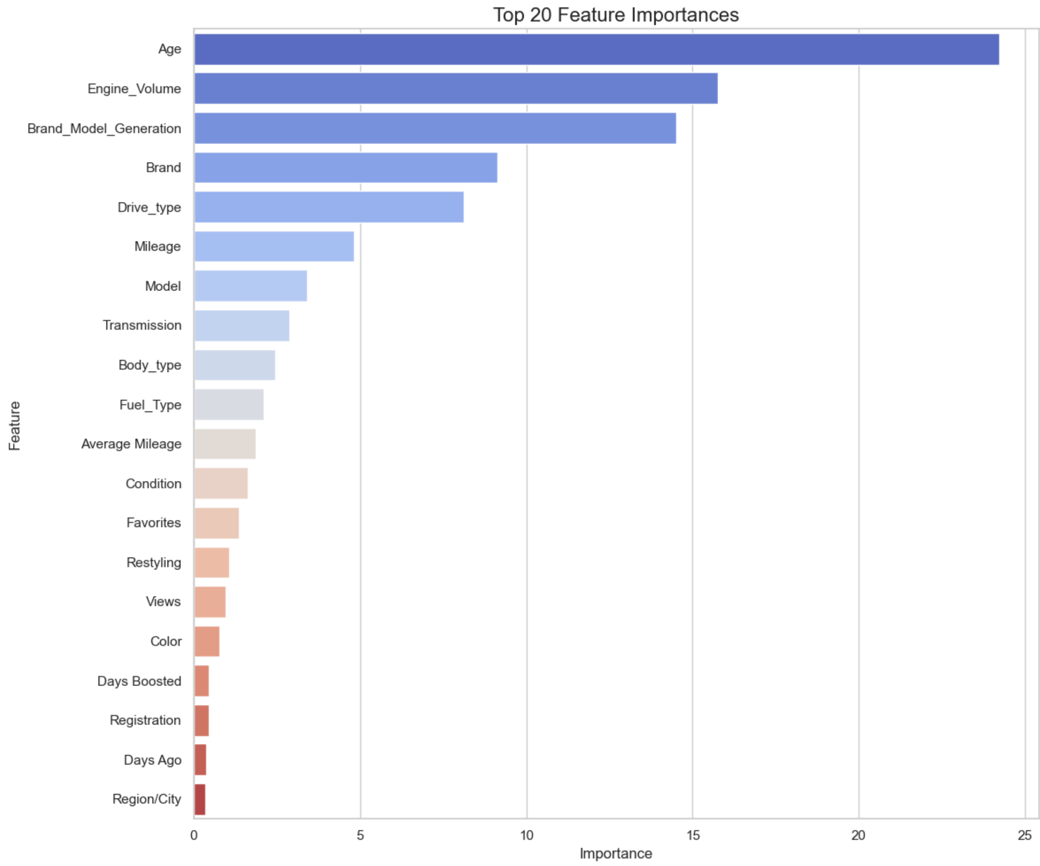

Figure 7. Top 20 features with highest impact on car price prediction using the CatBoost model.

- **Engine Volume:** Larger engine sizes often indicate more powerful or higher-class vehicles, which generally command premium pricing.
- **Brand Model Generation:** This feature captures generational and branding differences, reflecting buyer preferences for newer design iterations and brand reliability.
- **Brand and Model:** Both features directly relate to reputation, durability, and perceived luxury—key factors in consumer pricing decisions.
- **Drive Type and Transmission:** These technical specifications affect fuel efficiency, handling, and drivetrain performance.
- **Mileage and Average Mileage:** These metrics capture vehicle usage. Lower mileage is generally correlated with reduced wear, higher resale value, and greater buyer appeal.
- **Condition and Restyling:** Cars marked as "ideal" or recently restyled are typically priced higher due to better appearance or updated features.
- **User Engagement Features (Favorites, Views, Days Boosted):** These indirectly indicate demand and seller behavior, which may correlate with pricing expectations or listing quality.
- **Color, Registration, Region/City:** These variables likely capture aesthetic preferences, legal implications, or regional market variations, although their overall importance is lower.

These findings demonstrate that the CatBoost model relies on a blend of technical specifications, vehicle age, brand characteristics, and behavioral indicators to estimate car prices. This feature importance analysis adds transparency to the model's decision process and supports trust in its predictions.

4. Discussion

This study presented a data-driven model for predicting used car prices in the Bishkek market using a user-generated dataset from the *mashina.kg* platform. The CatBoost model demonstrated strong performance, particularly for mid-range vehicles priced below $60,000. The model achieved high accuracy across various metrics, including Mean Absolute Percentage Error (MAPE) and R^2 scores, confirming its effectiveness in predicting the prices of cars within the most commonly available price range in the dataset. Key features such as vehicle age, engine volume, and brand-model generation were found to be the most significant predictors, aligning well with established market valuation factors. These results suggest that, for the majority of vehicles, the model performs as expected, successfully capturing key pricing patterns based on well-established market principles.

However, the model's performance exhibited a noticeable decline for higher-priced vehicles, particularly those priced above $60,000. This drop in accuracy can primarily be attributed to the underrepresentation of premium vehicles in the dataset. The limited number of high-end car listings hindered the model's ability to generalize and predict accurately in this price range. This finding directly answers one of our research questions regarding the impact of data imbalance: the model struggles with higher-end vehicle predictions due to the lack of sufficient data. Thus, the study emphasizes the need for larger, more balanced datasets to improve prediction accuracy for underrepresented price segments.

Additionally, our brand-level analysis revealed that prediction errors were notably higher for brands with fewer listings. This further reinforces the challenges posed by data imbalance, particularly when it comes to underrepresented brands. The model's ability to predict prices accurately is influenced by the volume of available data for each brand, which highlights the necessity of improving data quality and quantity for brands with fewer listings. This supports the idea that model generalization improves with better data coverage across all car brands and price segments.

Another limitation that impacted the model's performance is the **manual data entry** process on the *mashina.kg* platform. Sellers often inflate car conditions to justify higher prices, especially in the "Condition" column, where cars are frequently labeled as "Ideal" or "Good" despite the actual condition varying significantly. This inconsistency in condition labeling introduces noise into the dataset, affecting the model's performance. This issue answers another research question: data quality, in particular, the reliability of user-generated data, plays a crucial role in model accuracy. Inconsistent data labeling undermines the effectiveness of the model, particularly when predicting prices for vehicles whose conditions have been inaccurately described.

The results of this study highlight the importance of consistent and accurate data labeling and the need for more comprehensive datasets that better represent premium vehicles and less common brands. To improve the model's robustness, future research should focus on acquiring more data for high-end vehicles, which would allow the model to better capture the pricing dynamics of these segments. Additionally, more advanced techniques such as **data augmentation** or **synthetic data generation** could help address the data imbalance, enriching the dataset with artificially generated entries for underrepresented price ranges and brands. Incorporating external variables, such as macroeconomic factors, seasonal demand fluctuations, and regional market trends, could also improve the model's predictive accuracy and generalizability.

In summary, while the model developed in this study provides a strong foundation for car price prediction in the Kyrgyz market, there is significant room for improvement. Addressing the limitations of data imbalance, enhancing data labeling consistency, and incorporating external factors will be critical steps toward refining the model and improving its predictive capabilities for all vehicle price ranges and brands. This research provides valuable insights into the use of machine learning for dynamic car pricing systems and sets the stage for future work in the development of more accurate and reliable pricing tools in the automotive market.

5. Conclusion

This study developed a machine learning model to predict used car prices in the Bishkek market, based on a user-generated dataset from the *mashina.kg* platform. Among several models tested, CatBoost emerged as the most effective, particularly for mid-range vehicles priced below $60,000. The model was able to successfully identify key pricing factors such as vehicle age, engine volume, and brand-model generation, which align with known market trends.

While the model performed well for mid-range vehicles, it struggled with high-end cars due to the underrepresentation of these vehicles in the dataset. The impact of **data imbalance** and **manual data entry** by sellers were key factors contributing to the model's limited generalizability for luxury brands.

Looking forward, this research lays the groundwork for building more robust pricing models. Future work should focus on expanding the dataset to include more high-end cars, improving data quality by standardizing feature labeling, and exploring external variables, such as economic indicators and market trends, to enhance predictive accuracy across all price ranges. This work not only contributes to the development of pricing tools for the Kyrgyz market but also provides a foundation for building dynamic car pricing systems in emerging markets globally.

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